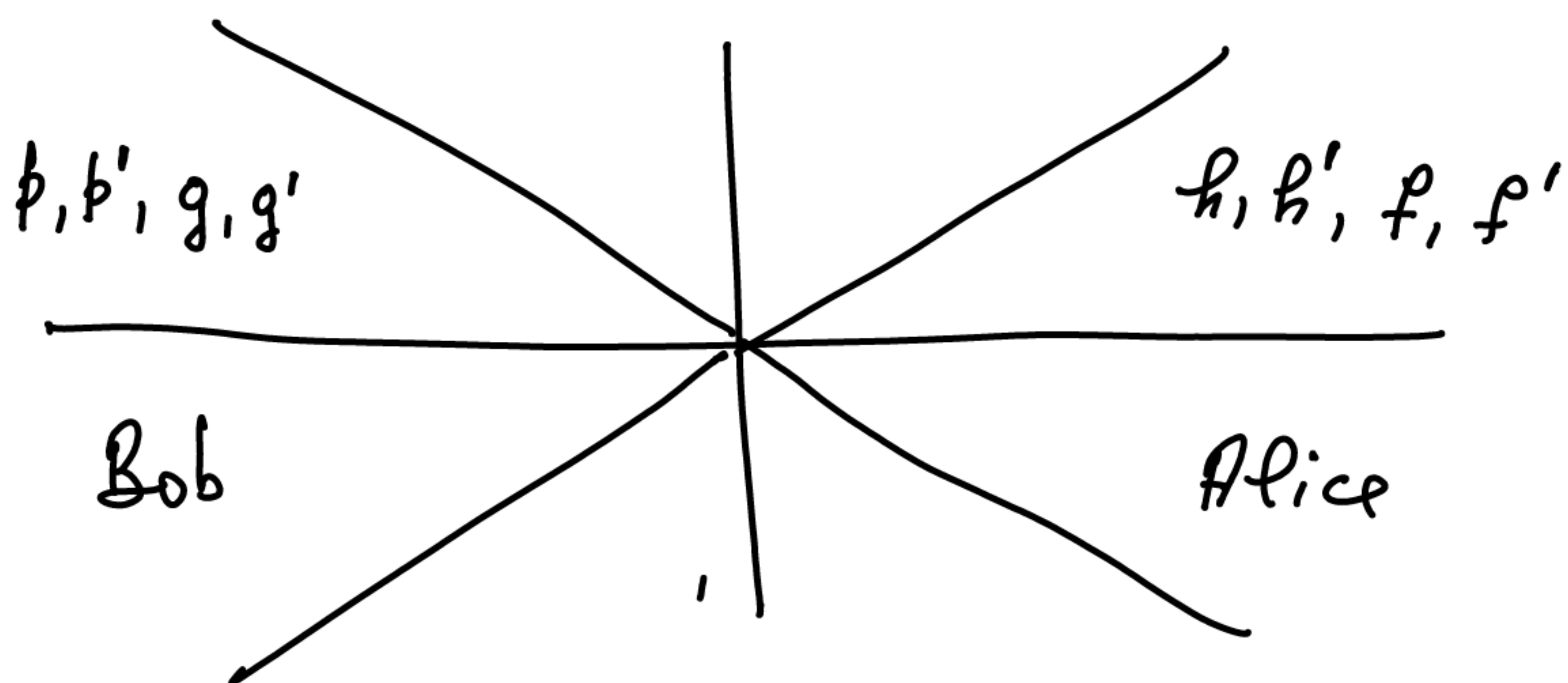


Majorana bilinear
operators from modular
localization.



$$\psi_f = a_f + a_f^\dagger \quad \text{massive Majorana}$$

$$\langle f | g \rangle = \int_{-\infty}^{\infty} d\theta \quad f^*(\theta) g(\theta)$$

$\downarrow$
 rapidity

$$\begin{cases} \delta^{1/2} f(\theta) = f(\theta + i\pi) \\ i f(\theta) = f^*(\theta) \end{cases}$$

We move to the dual space ω

$$f(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\theta e^{-i\omega\theta} f(\theta)$$

$$\left\{ \begin{array}{l} \delta^{1/2} f(\omega) = e^{-\pi\omega} f(\omega) \\ i f(\omega) = f^*(-\omega) \end{array} \right.$$

$$S = f^\dagger \delta^{1/2} \quad S f(\omega) = e^{\pi\omega} f^*(-\omega)$$

$$\langle f | g \rangle = \int_{-\infty}^{\infty} d\omega f^*(\omega) g(\omega)$$

Let us introduce Alice's operators

$$A = -i \frac{\psi(h) \psi(f)}{\|h\| \|f\|}$$

$$A' = -i \frac{\psi(h') \psi(f')}{\|h'\| \|f'\|}$$

One requires that

$$\operatorname{Re} \langle h | f \rangle = \operatorname{Re} \langle h' | f' \rangle = 0$$

From

$$\langle \psi_f, \psi_h \rangle = 2 \operatorname{Re} \langle f | h \rangle$$

$\Downarrow$

$$\langle \psi_f, \psi_h \rangle.$$

Thus

$$A^+ = \frac{i}{\|f\| \|h\|} \varphi(f) \varphi(h) = A$$

$$\begin{aligned} A^2 &= - \frac{1}{\|f\|^2 \|h\|^2} \varphi_h \varphi_f \varphi_h \varphi_f \\ &= \frac{1}{\|f\|^2} \frac{1}{\|h\|^2} \varphi_h^2 \varphi_f^2 = 1 \end{aligned}$$

(A, A') are Hermitian and dichotomic.

For the modular construction of

(f, f', h, h') we set

$$f = (1+s) \varphi(\omega) = \varphi(\omega) + e^{\pi\omega} \varphi(-\omega)$$

$$f' = (1+s) i \varphi(\omega) = i (\varphi(\omega) - e^{\pi\omega} \varphi(-\omega))$$

$$\varphi(\omega) \Rightarrow \text{real.}$$

For $(h, h') \Rightarrow$

$$h = (1+s) \underbrace{(\lambda + i\eta)}_{k(W_R) + i k(W_R)} \quad \lambda, \eta \text{ real}$$

$$h' = (1+s) (\lambda' + i\eta') \quad \lambda', \eta' \text{ real}$$

The conditions $\operatorname{Re} \langle f, h \rangle = \operatorname{Re} \langle f', h' \rangle = 0$

are implemented through Gram-Schmidt.

Example

$$\langle f|h \rangle = \int d\omega \left[\varphi(\omega) (1 + e^{-2\pi\omega}) + 2 \cosh \pi\omega \varphi(-\omega) \right] \lambda(\omega) \\ + \text{imaginary-terms}$$

||

$$\int d\omega \left[\varphi(\omega) (1 + e^{-2\pi\omega}) + 2 \cosh \pi\omega \varphi(-\omega) \right] \lambda(\omega) = 0$$

$\Downarrow$ GS $\hat{\varphi}(\omega)$

$$\lambda(\omega) = \sigma(\omega) - \frac{\langle \hat{\varphi} | \sigma \rangle}{\|\hat{\varphi}\|^2} \hat{\varphi}$$

$\downarrow$

free

$$\langle \hat{\varphi} | \lambda \rangle = \langle \hat{\varphi} | \sigma \rangle - \langle \hat{\varphi} | \sigma \rangle = 0$$

For Bob one proceeds in the same way

$\rightarrow$ twisted duality.

$$g = i(1 + s^+) \hat{g}(\omega)$$

$$g' = -i(1 + s^+) i \hat{g}(\omega)$$

$$p = i(1 + s^+) (\alpha(\omega) + i\beta(\omega))$$

$$p' = -i(1 + s^+) (\alpha'(\omega) + i\beta'(\omega))$$

$$\alpha, \beta, \alpha', \beta' \text{ real}$$

One imposes

$$\operatorname{Re} \langle g, p \rangle = \operatorname{Re} \langle g' / p' \rangle = 0$$

For the Bell-CHSH $\Rightarrow$

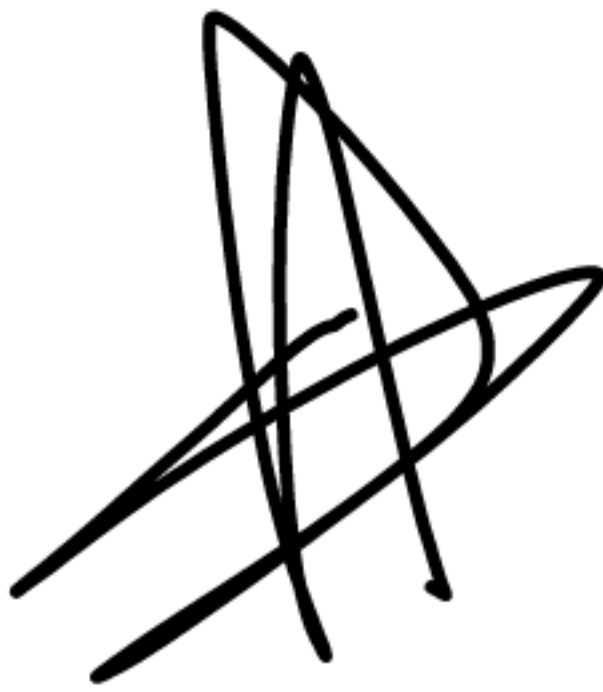
$$\langle C \rangle = \langle 0 | (A + A') B + (A - A') B' | 0 \rangle$$

$$[Alice, Bob] = 0$$

$$\langle AB \rangle = \frac{1}{\|h\| \|f\| \|b\| \|g\|} \left\{ \langle h|f \rangle \langle b|g \rangle + \langle f|b \rangle \langle h|g \rangle - \langle h|b \rangle \langle f|g \rangle \right\}$$

The numerical setup is done by means

{ Gaussian x Hermite polynomials
around $\omega \approx 0$



Numerical Study of Majorana Bilinears near the Tsirelson Bound

Appendix to the handwritten notes on modular localization

This appendix records the numerical setup associated with the clean handwritten construction in the preceding pages. The aim is to implement the modularly localized Majorana bilinears in a finite Hermite-Gaussian basis concentrated around modular frequency $\omega = 0$, impose the required real-orthogonality constraints by Gram-Schmidt, and evaluate the Bell-CHSH correlator.

1. Hermite-Gaussian finite basis

For a width $\sigma > 0$, the numerical trial space is generated by six Hermite-Gaussian modes ($n = 0, \dots, 5$):

$$\chi_n(\sigma)(\omega) = \exp[-\omega^2/(2\sigma^2)] H_n(\omega/\sigma) / \sqrt{2^n n!}.$$

The small-sigma regime isolates the modular spectral region ω approximately 0, which is the regime singled out in the last page of the notes.

2. Modular maps and localized vectors

The numerical implementation uses the ω -space Tomita maps written in the notes:

$$(s f)(\omega) = \exp(\pi i \omega) f^*(-\omega), (s^\dagger f)(\omega) = \exp(-\pi i \omega) f^*(-\omega).$$

Alice's vectors are generated with $(1+s)$. Bob's vectors are generated with the twisted-dual $(1+s^\dagger)$ construction. Independent Hermite-Gaussian seeds are allowed for the second Majorana factor in each bilinear.

3. Real Gram-Schmidt constraints

Hermiticity of each normalized bilinear requires the corresponding Majorana factors to anticommute. Numerically this is imposed through the real part of the one-particle scalar product. For example,

$$h \rightarrow h - [\text{Re} / \|f\|^2] f,$$

and similarly for the primed pair and for Bob. Thus the imposed conditions are $\text{Re} = \text{Re} = 0$ and $\text{Re} = \text{Re} = 0$. No additional condition requiring $\psi(g)$ and $\psi(g')$ to anticommute is imposed.

4. Bell-CHSH evaluation

For $A = -i \psi(h)\psi(f)/(\|h\| \|f\|)$ and the corresponding Bob bilinear, Wick contraction gives the four-point expression used in the handwritten notes. The CHSH combination is evaluated as

$$B_{\text{CHSH}} = + + - -.$$

The numerical search is performed over the finite Hermite-Gaussian coefficient vectors, with the Gram-Schmidt constraints enforced before evaluating the correlator.

5. Numerical convergence

sigma	B_CHSH	2 sqrt(2) - B_CHSH
1.0 x 10 ⁻²	2.82474588	3.68 x 10 ⁻³
5.0 x 10 ⁻³	2.82718767	1.24 x 10 ⁻³
2.0 x 10 ⁻³	2.82820789	2.19 x 10 ⁻⁴
1.0 x 10 ⁻³	2.82837085	5.63 x 10 ⁻⁵

For reference, $2 \sqrt{2} = 2.828427124746\dots$. At $\sigma = 10^{-3}$ the reported optimized value is

$$\mathbf{B_CHSH = 2.8283708483,}$$

corresponding to $B_CHSH/(2 \sqrt{2})$ approximately 0.9999801. Thus the finite six-mode trial space reaches about 99.998% of the Tsirelson value.

6. Four correlators at $\sigma = 10^{-3}$

correlator	value	Tsirelson target
$\langle AB \rangle$	0.7071076745	$+1/\sqrt{2}$
$\langle A'B \rangle$	0.7070902621	$+1/\sqrt{2}$
$\langle AB' \rangle$	0.7071025976	$+1/\sqrt{2}$
$\langle A'B' \rangle$	-0.7070703141	$-1/\sqrt{2}$

The four entries approach the ideal CHSH pattern ($1/\sqrt{2}$, $1/\sqrt{2}$, $1/\sqrt{2}$, $-1/\sqrt{2}$). This is a stronger diagnostic than the CHSH sum alone: each individual correlation approaches its Tsirelson configuration.

7. Numerical stability and interpretation

The $\sigma = 10^{-3}$ result was reported stable under increasing the Gauss-Hermite quadrature from 120 to 180 and 240 points, yielding 2.828370848290538 at the displayed precision. Within this numerical experiment, the trend is therefore consistent with

$$\sigma \rightarrow 0 \Rightarrow \omega \rightarrow 0 \Rightarrow \mathbf{B_CHSH \rightarrow 2 \sqrt{2}.$$

Important status note. The numbers in this appendix record the numerical results obtained for the finite Hermite-Gaussian implementation described above. They are numerical evidence for near saturation; by themselves they are not an analytic proof that the limiting value equals $2 \sqrt{2}$. An analytic control of the $\sigma \rightarrow 0$ limit would be a separate step.